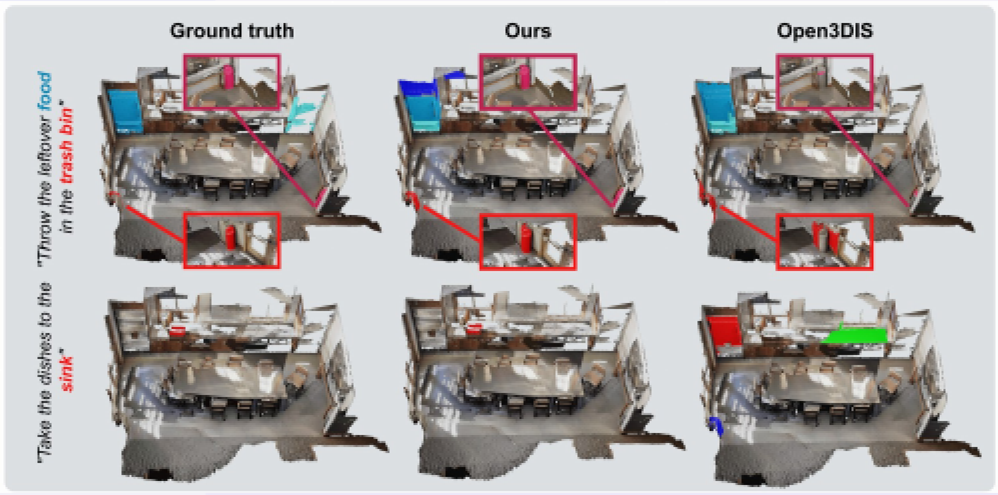
Open-YOLO 3D: Towards Fast and Accurate Open-Vocabulary 3D Instance Segmentation

Mohamed El Amine Boudjoghra, Angela Dai, Jean Lahoud, Hisham Cholakkal, Rao Muhammad Anwer, Salman Khan, Fahad Shahbaz Khan

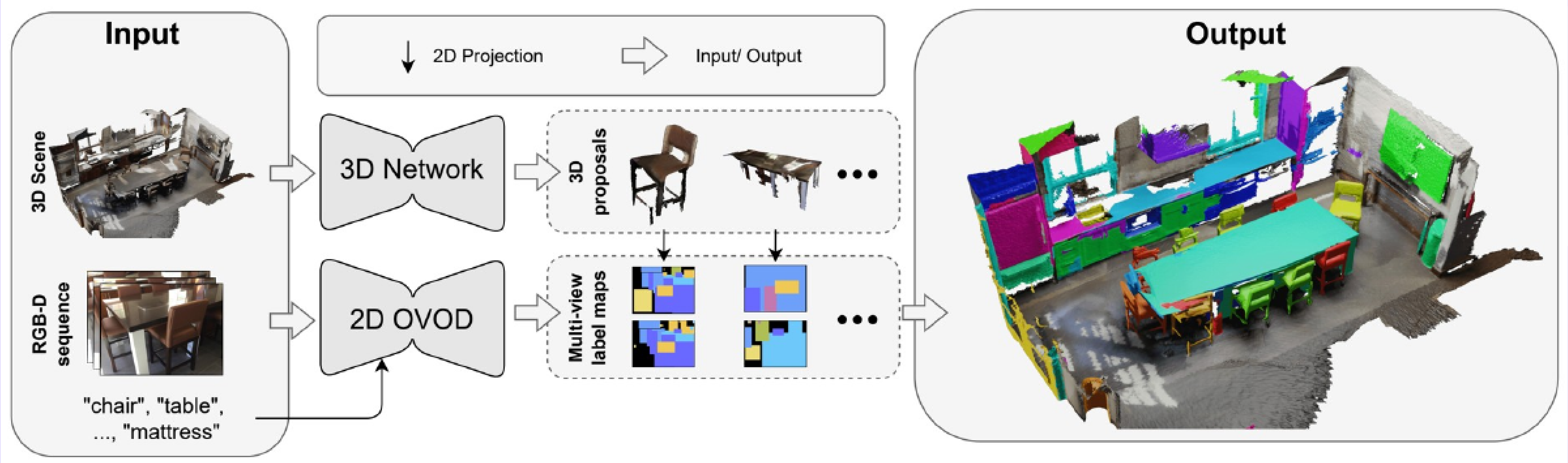
MBZUAI, TUM, Aalto University, ANU, Linköping University

01 Open-YOLO 3D: Paper Title and Authors

- Authors and affiliations are listed for all contributors.
  - MBZUAI, TUM, Aalto Univ., ANU, Linköping Univ.
- Right: qualitative comparison — ours vs baselines on ScanNet200; shows cleaner 3D masks.
- Collaborating institutions demonstrate broad international reach.

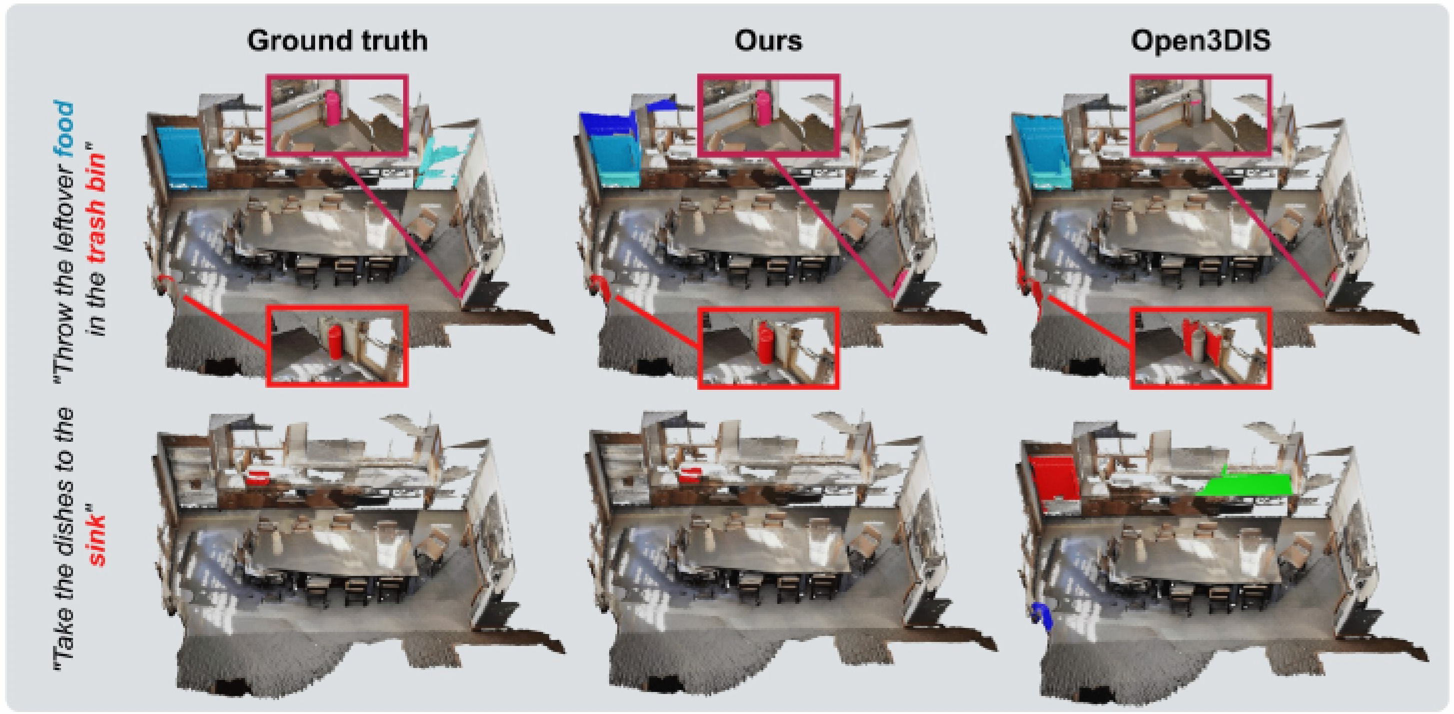


02 Open-YOLO 3D at a Glance (Visual)



- **Inputs** → OV boxes + 3D masks.
- **3D points** → LG maps via projection.
- **MVPDist** aggregates votes → **final class**.

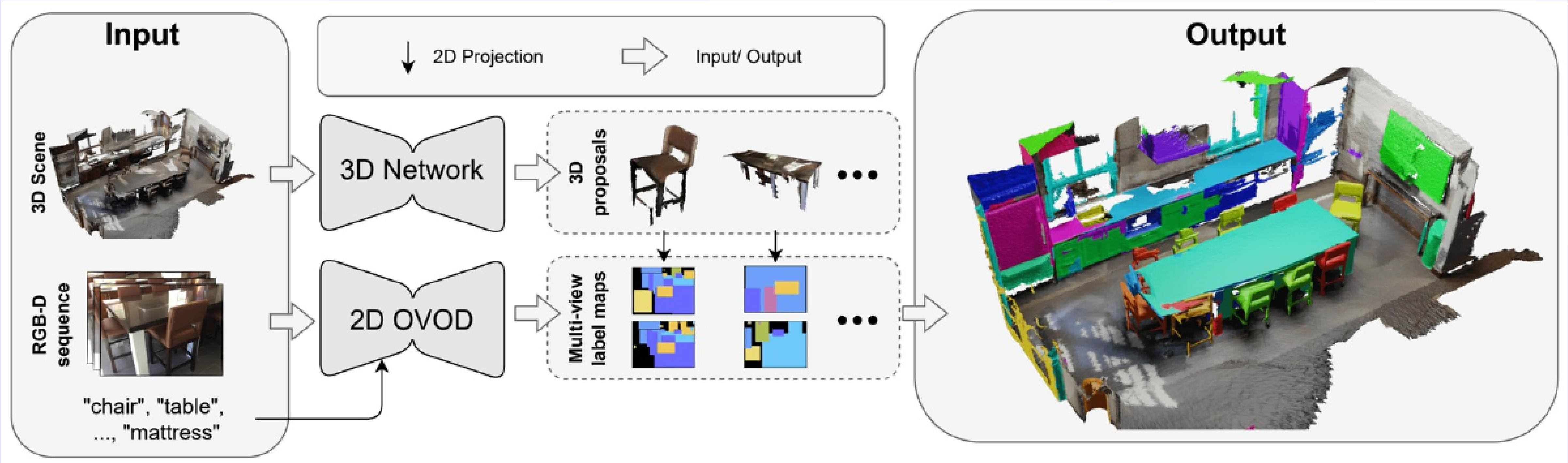
03 Open-YOLO 3D: Fast Open-Vocabulary 3D Instance Segmentation



- Class-agnostic 3D proposals are generated without 2D mask dependence.
  - Class-agnostic 3D proposals; no SAM
  - Label masks via 2D OV detection features
- Two evaluation scenarios are supported for comprehensive analysis.
  - Oracle masks
- Results yield state-of-the-art accuracy with substantial speedups.
  - SOTA on ScanNet200 & Replica
  - ~16x speedup; 24.7% mAP at 22 s/scene

04

Introduction: Fast Open-Vocabulary 3D Instance Segmentation with Open-YOLO 3D

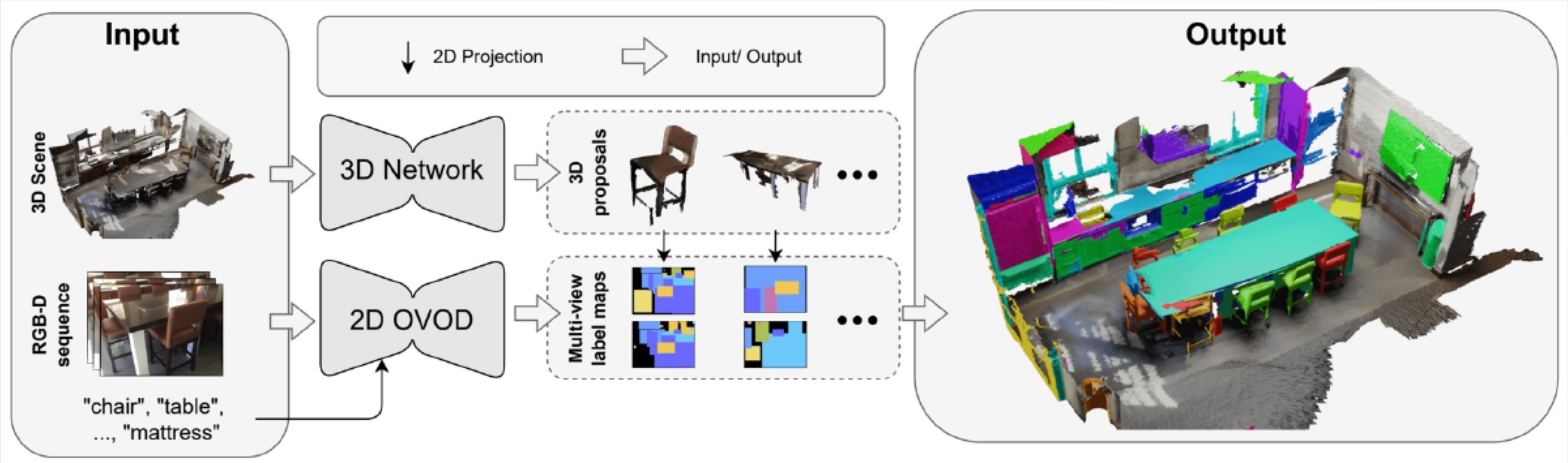


- Open-vocabulary 3D segmentation requires higher throughput for practical use.
  - OV-3D needs speed; SAM/CLIP lifting is slow
- Open-vocabulary 2D detection and low-granularity label maps replace heavy lifting.
  - OV 2D detection + LG label maps + 2D–3D projection
- Key contributions deliver faster inference with strong accuracy.
  - Detection-based labeling
  - Box-only scoring via MVPDist
  - SOTA speed–accuracy

[34, 42, 23, 55]

05

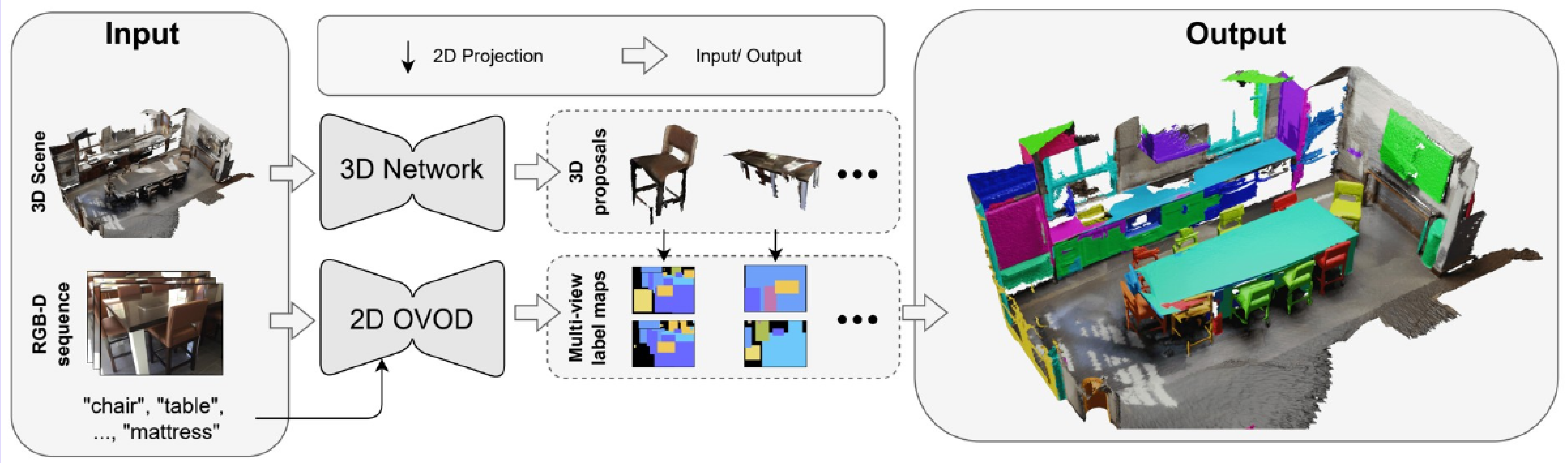
Related Work: From Closed- to Open-Vocabulary 3D Segmentation



- Closed-vocabulary 3D methods span bottom-up grouping and top-down proposals.
  - Bottom-up, top-down, transformers
  - Callout (legend): blue blocks = 2D OVOD; green masks = 3D proposals
- Open-vocabulary 2D recognition advances enable strong zero-shot capability.
  - OVOD, OVSS, OVIS
- Open-vocabulary 3D methods lift 2D cues but often incur heavy latency.
  - OpenScene, ConceptGraphs, OpenMask3D, OVIR-3D, Open3DIS
- A persistent gap remains due to SAM and CLIP dependence causing slow pipelines.
  - SAM/CLIP heavy; slow

06

Preliminaries: OV 3D Instance Segmentation and Baseline Pipeline

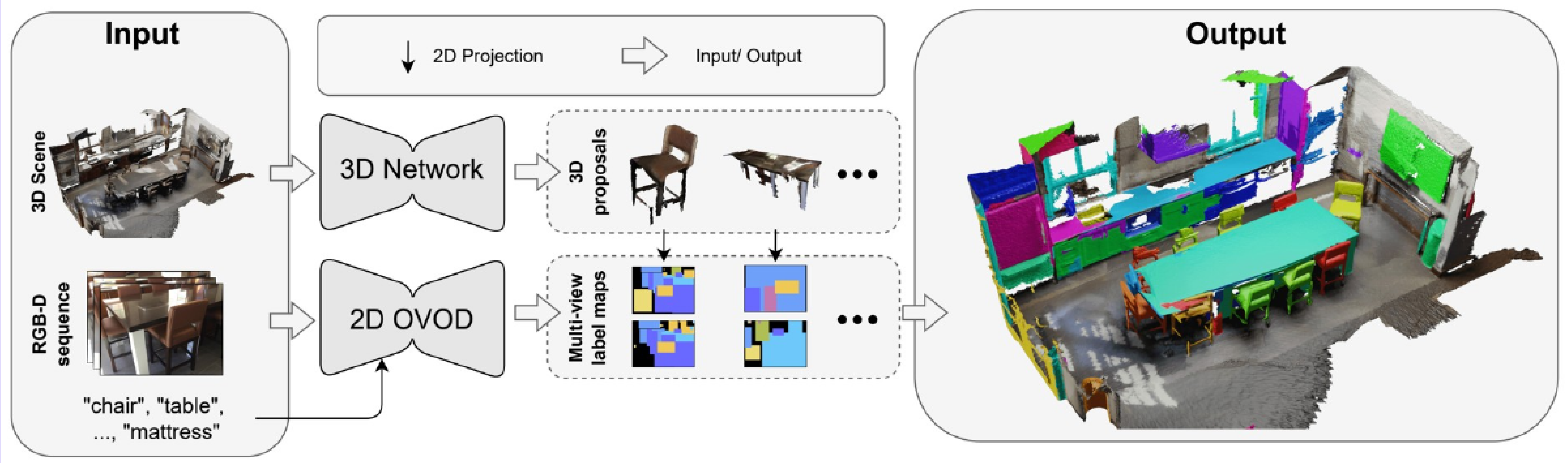


- The task outputs point-cloud instance masks with open-set class labels.
  - Point-cloud masks with open-set labels
- A baseline uses 3D proposals with SAM and CLIP for mask feature lifting.
  - 3D proposals + SAM/CLIP mask features
  - Callout: **baseline** flow — orange arrows; gray = SAM/CLIP lifting
- The baseline exhibits long runtimes due to 2D mask operations and lifting.
  - 5–10 min/scene due to 2D mask ops and lifting

[42, 39, 23, 55]

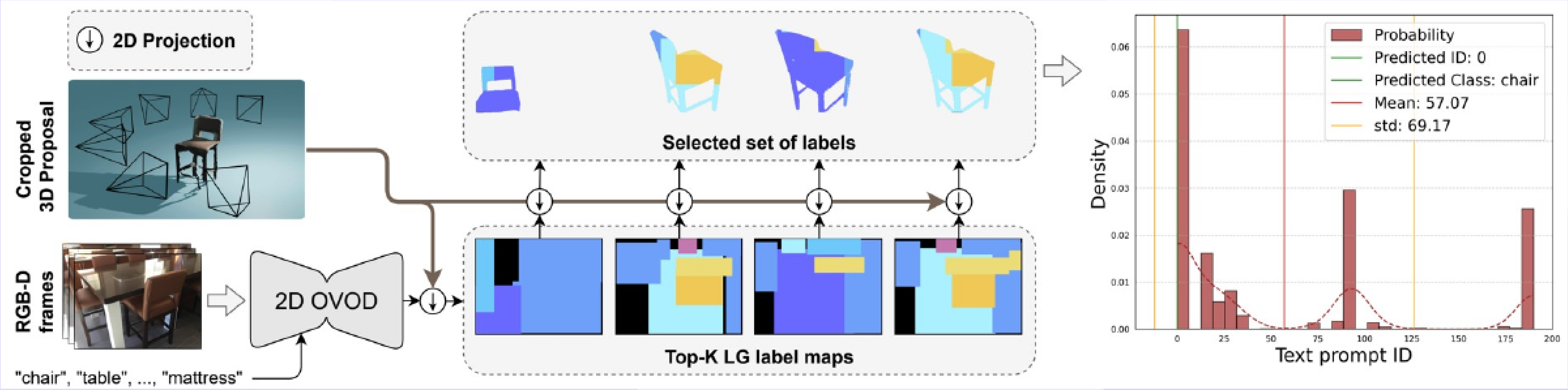
07

Method Overview (Textual)



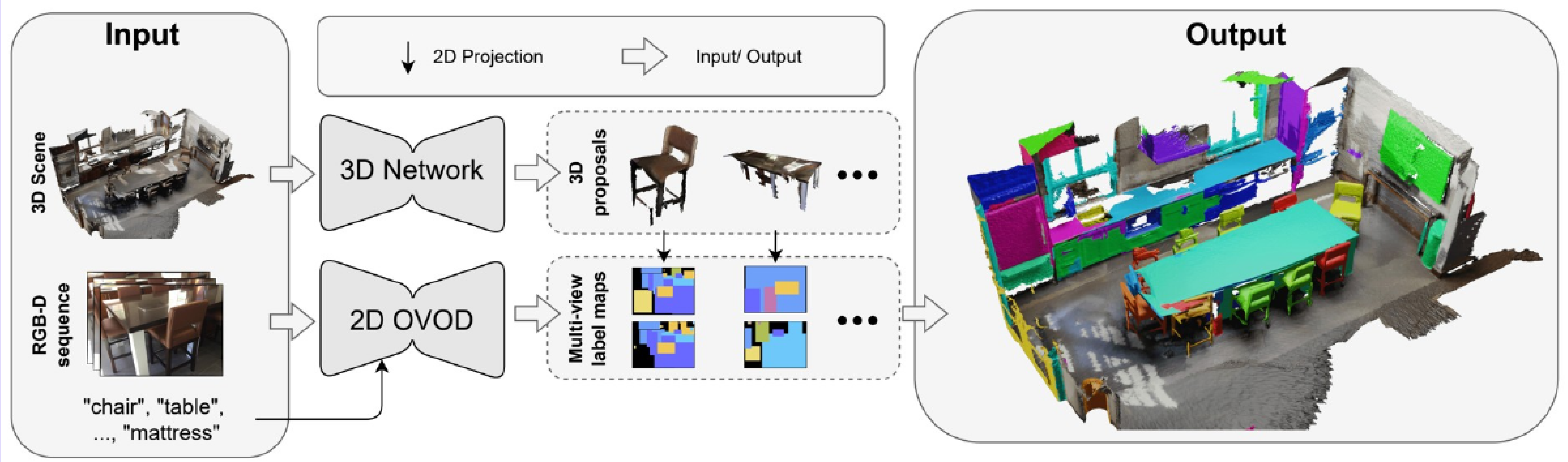
- Open-vocabulary 2D detection assigns labels to 3D masks without 2D segmentation.
  - Callout: flow — blue detector labels green 3D masks via LG maps
  - Avoids 2D segmentation and CLIP lifting
- Parallel visibility computation accelerates multi-view reasoning on the GPU.
  - Parallel batched ops
- The overall design targets fast inference with strong scalability.

Pipeline Steps: From Proposals to MVPDist



- Class-agnostic 3D proposals are generated and views provide open-vocabulary boxes.
  - 3D proposals → OV boxes → LG maps
  - Batched projection/visibility
  - Top-k frames → MVPDist → label
- Top-k visible frames are selected per mask for robust label aggregation.
- Multi-View Prompt Distribution assigns the final class with **majority support**.

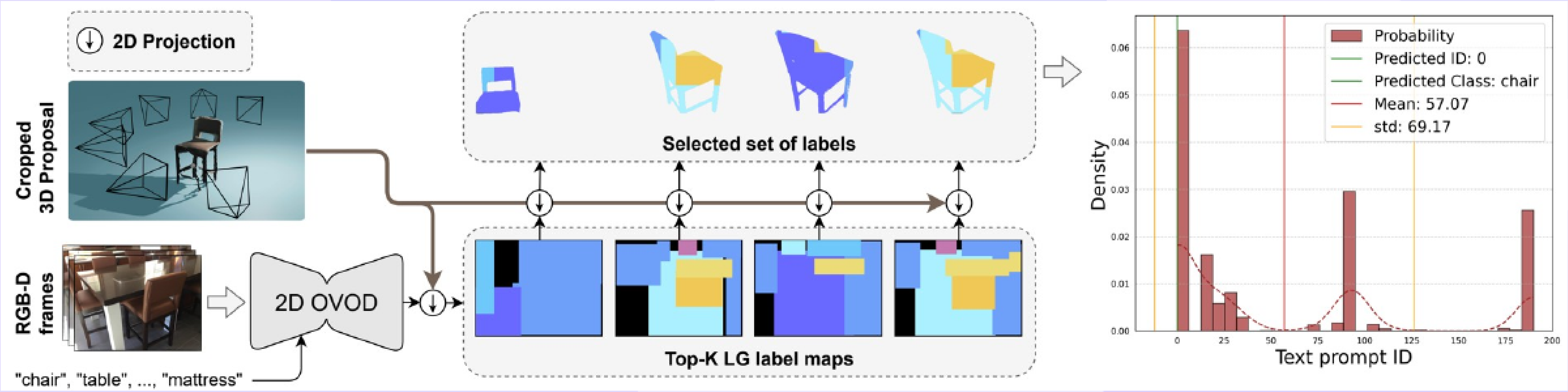
3D Proposal Generation (Mask3D)



- Mask3D produces fast class-agnostic proposals with a transformer decoder.
  - 3D CNN + transformer decoder
  - Class-agnostic masks
- The proposal stage is compatible with other **3D instance backbones**.
- The proposals enable efficient downstream labeling without retraining.

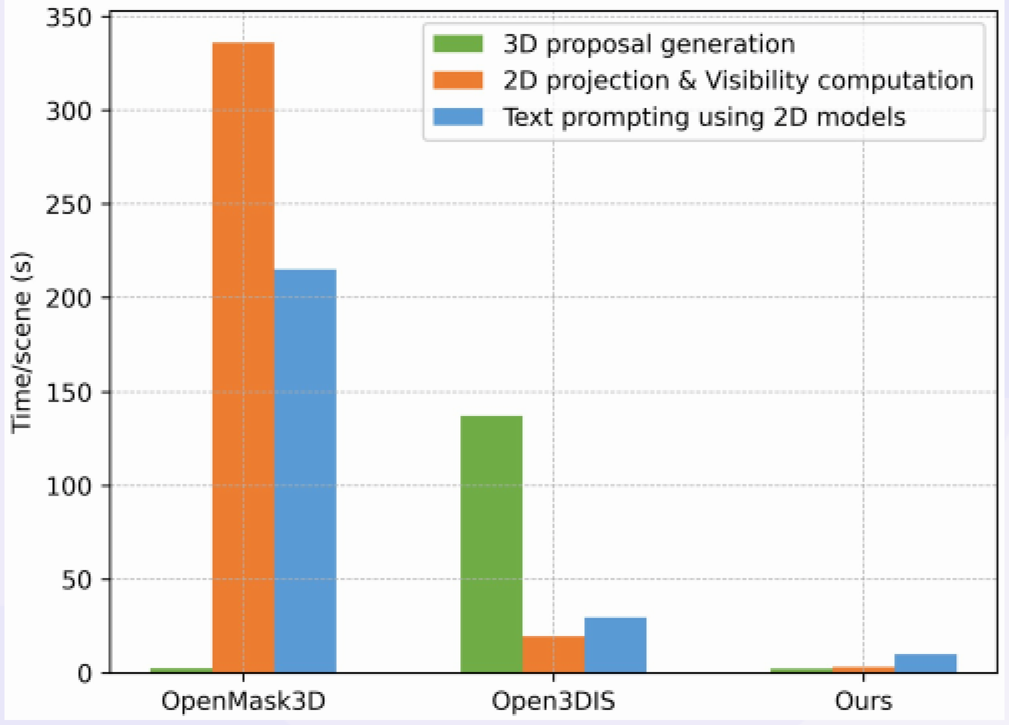
[39]

Low-Granularity Label Maps (YOLO-World)



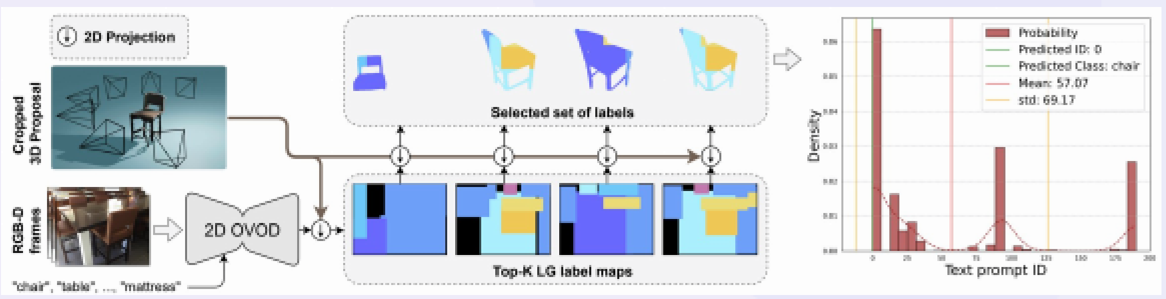
- Open-vocabulary detections create low-granularity label maps per frame.
  - Fill by size-sorted boxes
  - Favor larger/near objects
- Box ordering by area prioritizes **dominant visible objects**.
- Label maps enable fast label lookup during projection.

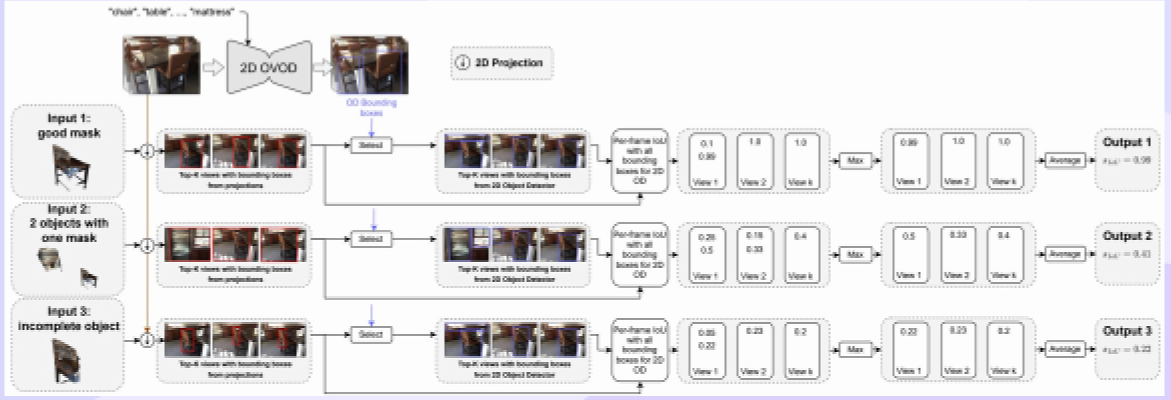
Accelerated Visibility (VAcc)



- Batched projection computes in-frame and occlusion visibility for all points.
  - Compute V_f, V_d, per-mask visibility
- Single-shot GPU operations deliver large runtime savings.
  - Single-shot GPU ops
  - Legend: **blue**=ours, gray=baseline
- Per-mask visibility guides **top-k frame selection** reliably.

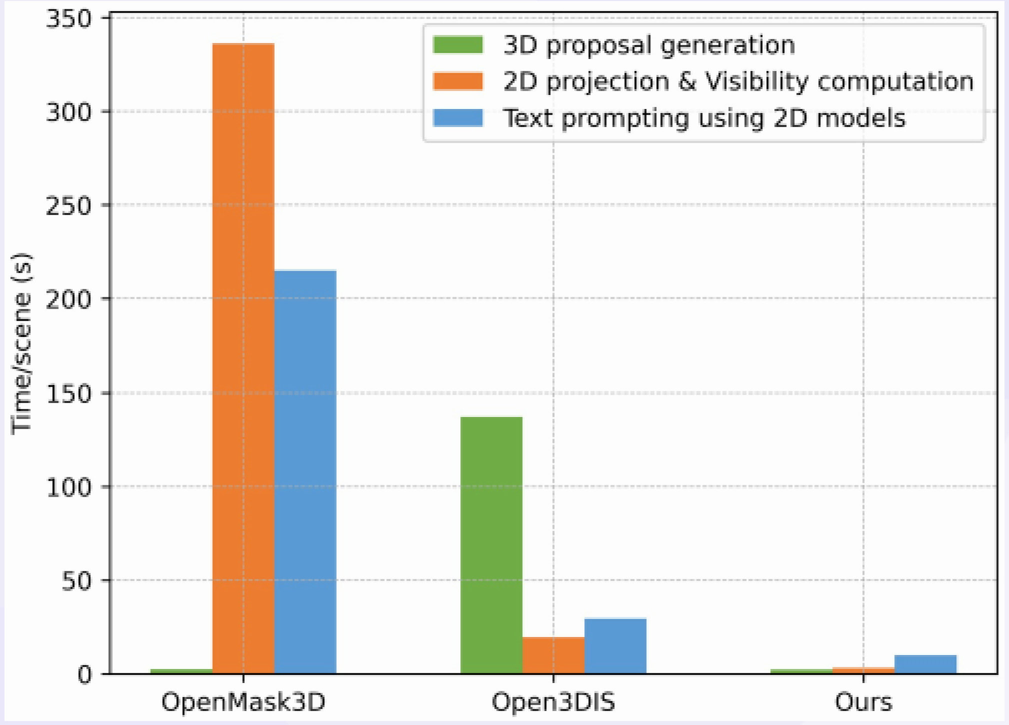
12 Multi-View Prompt Distribution and Confidence Scoring





- Left: per-view votes form **MVPDist** histogram → class.
- Right: s_m combines IoU with class prob for ranking.

13 Experimental Setup: Datasets, Metrics, and Implementation



- Experiments evaluate ScanNet200 and Replica with standard metrics.
  - ScanNet200, Replica; AP@.25/.50, mAP
  - Legend: **blue**=ours, gray=baseline (plots)
- Implementation uses YOLO-World for 2D priors and Mask3D for proposals.
  - YOLO-World (XL) + Mask3D; A100 40GB
- Frame sampling follows prior work to ensure **fair comparisons**.

14 Main Results on ScanNet**200**

Method	3D proposals from 2D masks	SAM for 3D mask labeling	mAP	mAP50	mAP25	head	comm	tail	time/scene (s)
Mask3D [39] (Closed Vocab.)	×	×	26.9	36.2	41.4	39.8	21.7	17.9	13.41
SAM3D [50]	✓	×	6.1	14.2	21.3	7.0	6.2	4.6	482.60
OVIR-3D	✓	×	13.0	24.9	32.3	14.4	12.7	11.7	466.80
Open3DIS [34]	✓	×	23.7	29.4	32.8	27.8	21.2	21.8	360.12
OpenScene [35] (2D Fusion)	×	×	11.7	15.2	17.8	13.4	11.6	9.9	46.45
OpenScene [35] (3D Distill)	×	×	4.8	6.2	7.2	10.6	2.6	0.7	**0.26**
OpenScene [35] (2D-3D Ens.)	×	×	5.3	6.7	8.1	11.0	3.2	1.1	46.78
OpenMask3D [42]	×	✓	15.4	19.9	23.1	17.1	14.1	14.9	553.87
Open3DIS [34]	×	×	18.6	23.1	27.3	24.7	16.9	13.3	57.68
Open-YOLO 3D (Ours)	×	×	**24.7**	**31.7**	**36.2**	**27.8**	**24.3**	**21.6**	21.8

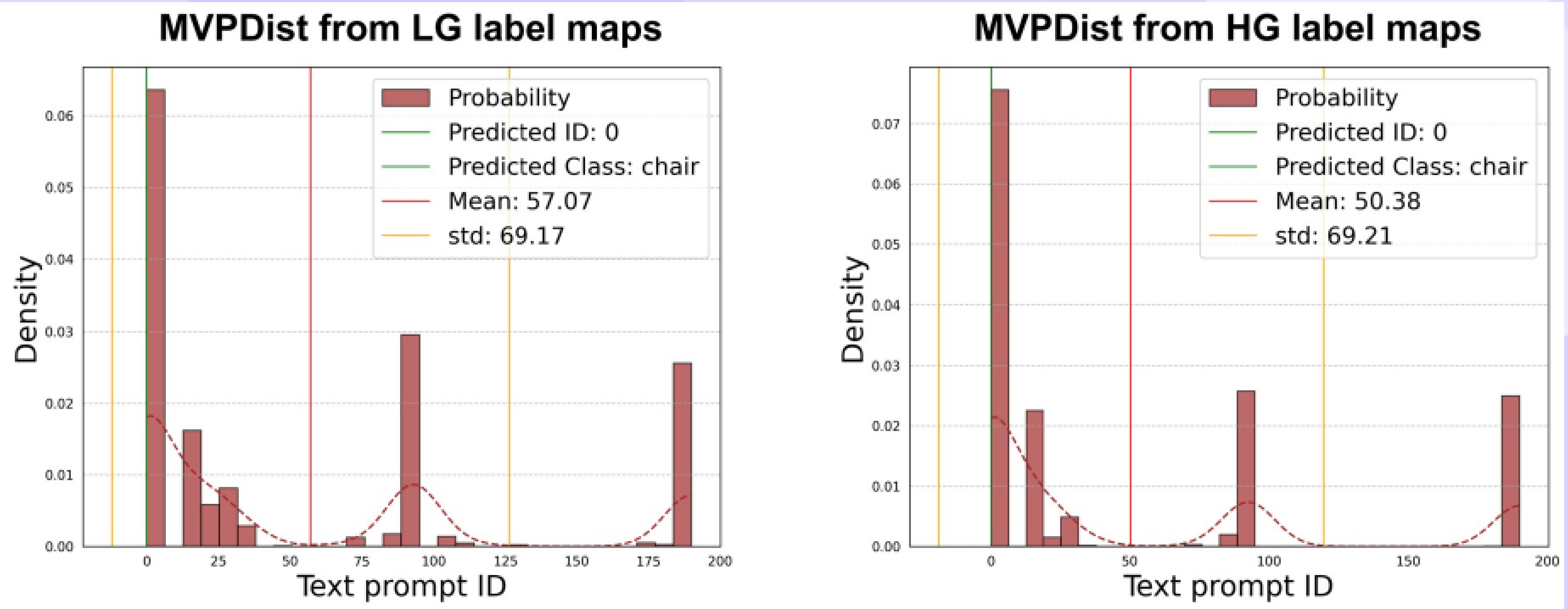
- Open-YOLO 3D achieves state-of-the-art mAP using only 3D proposals.
  - mAP 24.7; 21.8 s/scene
  - Interpretation: mAP50 31.7 at 21.8 s/scene — SOTA trade-off
- Oracle-mask evaluation shows strong gains over CLIP-based baselines.
  - MVPDist 39.6 mAP; > CLIP-based
- Read first: mAP50 = primary accuracy; time/scene = efficiency; head/common/tail = frequency bins.

15 Generalization to Replica

Method	3D proposals from 2D masks	SAM for 3D mask labeling	mAP	mAP50	mAP25	time/scene (s)
OVIR-3D	✓	×	11.1	20.5	27.5	52.74
Open3DIS [34]	✓	×	18.5	24.5	28.2	187.97
OpenScene [35] (2D fusion)	×	×	10.9	15.6	17.3	317.327
OpenScene [35] (3D Distill)	×	×	8.2	10.5	12.6	**04.29**
OpenScene [35] (2D-3D Ens.)	×	×	8.2	10.4	13.3	320.127
OpenMask3D [42]	×	✓	13.1	18.4	24.2	547.32
Open3DIS [34]	×	×	14.9	18.8	23.6	35.08
Open-YOLO 3D (Ours)	×	×	**23.7**	**28.6**	**34.8**	16.6

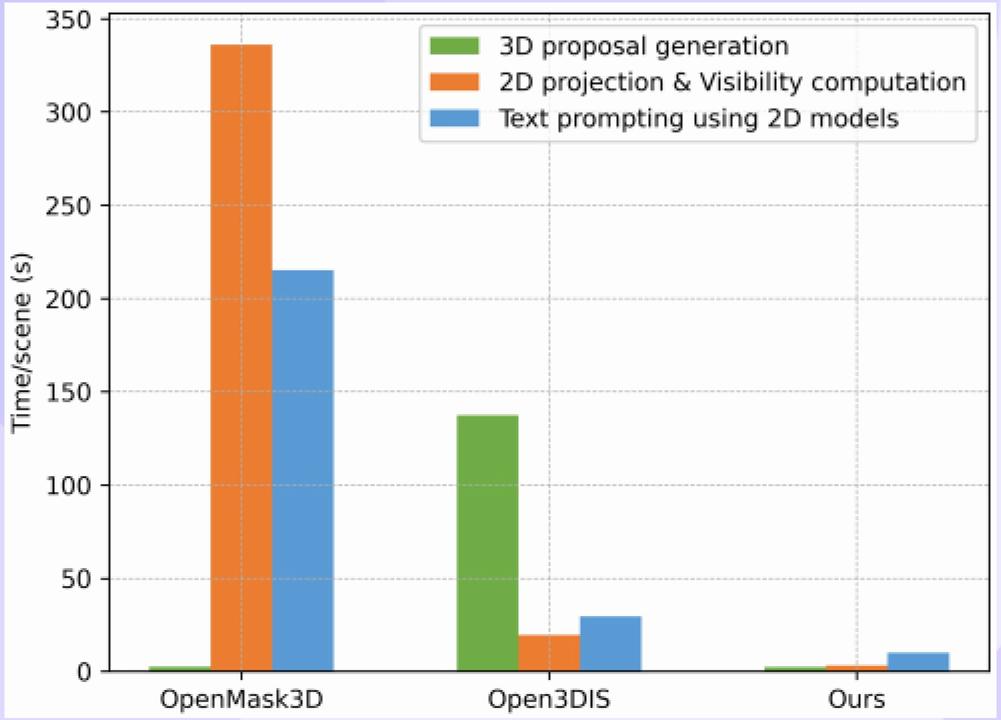
- Cross-dataset tests confirm strong **transfer** with lower runtime.
  - Ours 23.7 mAP; 16.6 s/scene
  - Interpretation: +8.8 mAP over Open3DIS at much lower runtime
- The approach is significantly faster than CLIP-based alternatives.
  - ~11x faster than CLIP-based SOTA
- Results highlight robustness across **indoor benchmarks**.

16 Ablations: Design Choices and Top-K Views



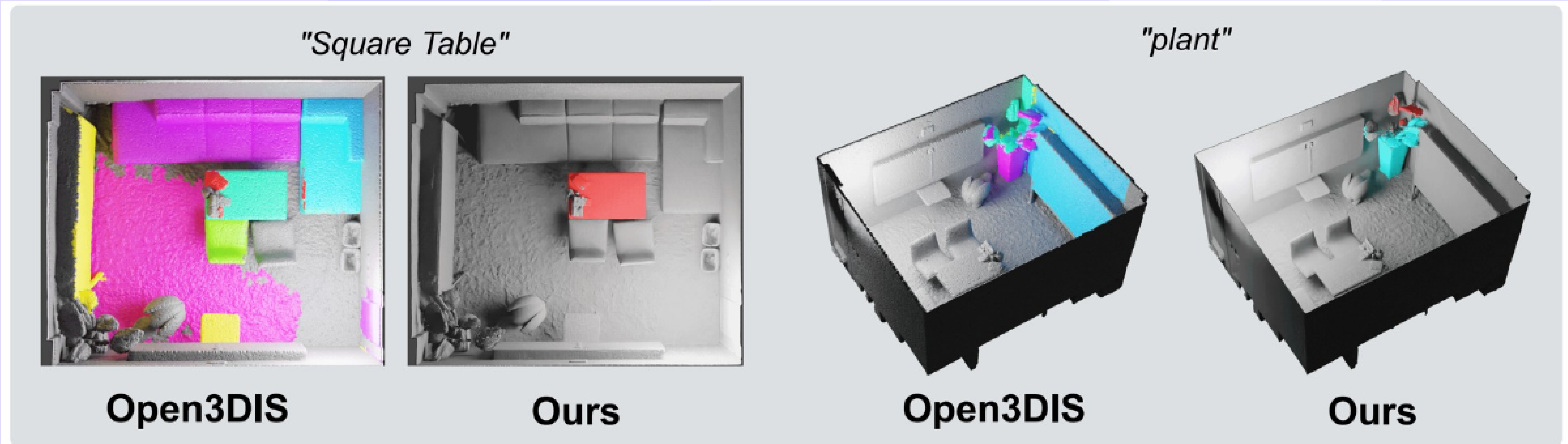
- Replacing SAM and CLIP with MVPDist and YOLO-World improves accuracy.
  - MVPDist + YOLO-World boosts mAP
- GPU-accelerated visibility reduces runtime with no mAP drop.
  - 376.4 s → 17.9 s at same mAP
  - Interpretation: **VAcc** drives the large speedup
- Low-granularity maps provide similar accuracy with much faster inference.
  - LG ~5x faster; similar mAP

17 Efficiency: Runtime and Speedups



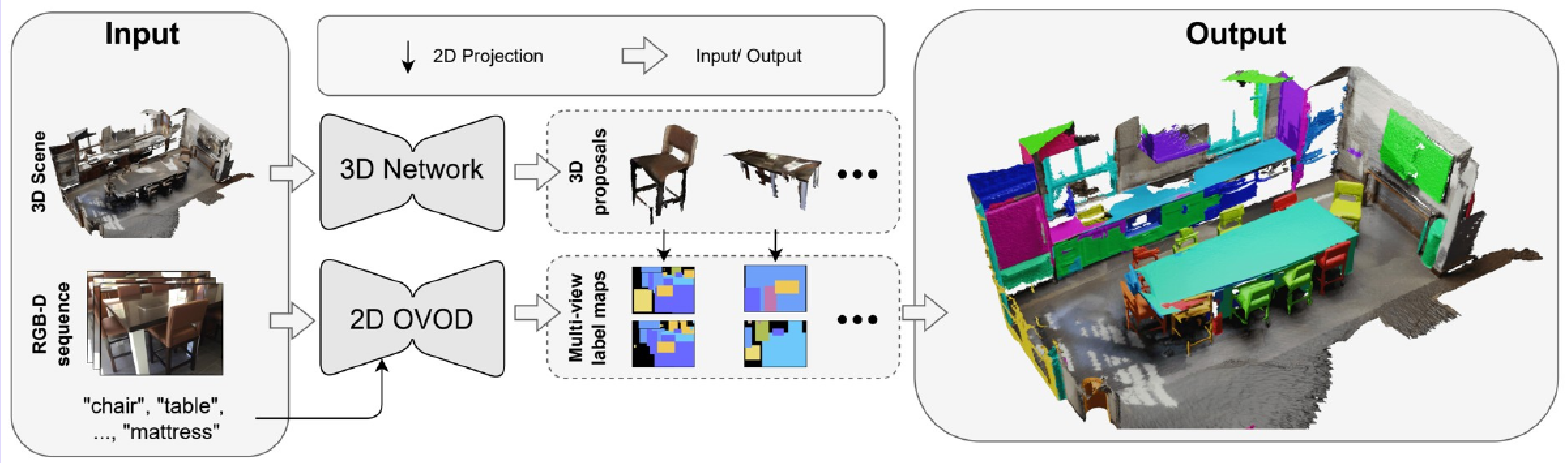
- End-to-end runtime is substantially reduced on both datasets.
  - ScanNet200: 21.8 s/scene
  - Replica: 16.6 s/scene
- Relative speedups over recent methods are consistently large.
  - ~16x vs Open3DIS
- Time breakdown attributes gains to **VAcc and LG maps**.

18 Qualitative Insights and Limitations



- Qualitative comparisons show higher precision than Open3DIS in Replica.
  - Higher precision than Open3DIS
- The approach depends on 3D proposal quality for small and thin objects.
  - Depends on 3D proposals; small objects
- Future extensions could leverage **fast 2D segmentation** to boost recall.

19 Conclusion and Key Takeaways



- **Takeaway: Open-YOLO 3D is a fast, accurate OV-3D pipeline without SAM/CLIP.**
- Open-vocabulary detection and MVPDist enable fast and accurate OV-3D.
  - OV detection + MVPDist = fast, accurate OV-3D
- The pipeline removes SAM and CLIP dependencies for simpler deployment.
  - No SAM/CLIP; 20–22 s/scene
- The method achieves a leading speed–accuracy trade-off across datasets.
  - SOTA accuracy–speed on ScanNet200/Replica

Thank You

Questions & Discussion